

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Based Insurance Claim Fraud Detection and Risk Assessment System

Abstract

Insurance fraud is a major challenge for insurance companies because fraudulent claims can result in financial losses and increase the time and effort required for claim verification. This project proposes an AI-Based Insurance Claim Fraud Detection and Risk Assessment System that uses machine learning to analyze insurance claim data and identify claims that may require further investigation. The system processes claim information through data preprocessing, missing-value handling, feature engineering, and class-imbalance handling, followed by the training and evaluation of multiple machine learning models to select an effective fraud detection model. When a new claim is submitted, the trained model analyzes the available claim and policy-related features and generates a fraud probability, which is further used to calculate a risk score and risk level such as Low, Medium, High, or Critical. To make the prediction understandable, the system integrates Explainable Artificial Intelligence (XAI) using SHAP, which displays the important factors contributing to the prediction for each claim. High-risk claims are automatically prioritized for investigation, where authorized investigators can review claim details, examine the AI prediction and explanation, add investigation notes, update the investigation status, and record the final decision. The system also provides secure user authentication, role-based access control, customer and policy management, claim management, fraud detection, risk assessment, explainable AI, investigation management, notifications, dashboards, analytics, model performance evaluation, report generation, and audit logging. The machine learning models are evaluated using precision, recall, F1-score, ROC-AUC, and confusion matrix to ensure that the selected model provides reliable fraud-detection performance. By combining AI-based claim analysis, risk-based prioritization, explainable predictions, and a structured investigation workflow, the proposed system provides an efficient and transparent platform to support insurance claim verification and fraud investigation.

Keywords: Insurance Fraud Detection, Insurance Claims, Machine Learning, Risk Assessment, Fraud Prediction, Explainable AI, SHAP, Class Imbalance, Claim Analysis, Fraud Risk Scoring

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References (Category specific)

- 1.Ahmed A. Khalil, Zaiming Liu, Ahmed Fathalla, Ahmed Ali, Ahmad Salah, **“Machine Learning Based Method for Insurance Fraud Detection on Class Imbalance Datasets With Missing Values,”** *IEEE Access*, Vol. 12, 2024, pp. 155451–155468, DOI: 10.1109/ACCESS.2024.3468993.
- 2.Itri Bouzgarne, Mohamed Youssfi, Qbadou Mohammed, Omar Bouattane, **“Performance Comparative Study of Machine Learning Algorithms for Automobile Insurance Fraud Detection,”** *2019 Third International Conference on Intelligent Computing in Data Sciences (ICDS)*, 2019, pp. 1–4, DOI: 10.1109/ICDS47004.2019.8942277.
- 3.Ahmed Urunkar, Anjali Khot, Rohan Bhat, N. Mudegol, **“Fraud Detection and Analysis for Insurance Claim Using Machine Learning,”** *2022 IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems (SPICES)*, 2022, pp. 406–411, DOI: 10.1109/SPICES52834.2022.9774071.
- 4.Kiet Ha, Lucas Stowe, Chandranil Chakrabortii, **“Improving Insurance Fraud Detection With Generated Data,”** *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, 2024, pp. 2008–2013, DOI: 10.1109/COMPSAC61105.2024.00321

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
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Project Guide

Name: Mrs. K. Deepika

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.